Real-Time Front-End Adaptation for Energy-Efficient mmWave Radar Sensor

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***Abstract*—Radar sensor systems often suffer from suboptimal performance in dynamic environments due to static front-end configurations. While cognitive radar approaches have been proposed to address this issue through adaptive processing, they are often computationally intensive and unsuitable for real-time deployment. This work proposed a lightweight, feedback-driven adaptation framework on a frame-by-frame basis. Target information is extracted from the Range-Doppler map and detection accuracy is calculated by range errors between predicted and detected values. A pre-trained machine learning (ML) model adapts the front-end with low latency, while risk levels of errors are assessed using predefined policies. Proposed work was validated on an AWR1843 mmWave radar sensor, achieves a $2.7\times$ improvement in energy efficiency while satisfying the latency constraints.**

***Index Terms*—Adaptive Configuration, Closed-loop Control, mmWave Radar Sensor, Real-time Processing.**

I. Introduction

Integrating AI and edge computing has enabled intelligent sensing and computation, which are crucial for autonomous systems and next-generation wireless networks [1]. However, traditional sensor architectures remain limited by static configurations, making them inflexible in dynamic environments.

Cognitive sensing addresses this challenge via adaptive feedback to dynamically reconfigure system parameters [2]. Table I shows the comparison of prior works, [3] demonstrate spectral-aware waveform adaptation in closed-loop system but suffer from high computational cost. [4]–[6] propose compressed sensing, efficient FFT-based waveform design methods with stopband and correlation constraints; however, fall short in performance in reality or incur high complexity during fine-tuning. [7], [8] use look-up table (LUT) based adaptation with limited flexibility.

Methods	closed-loop system	Adaptation Latency	Adaptation Flexibility	Computational cost
Waveform adaptation [3]	O	↑	↓	↑
Signal Processing Based [4]–[6]	X	↑	↑	↑
LUT based [7], [8]	O	↓	↓	↓
Proposed work	O	↓	↑	↓

TABLE I
Comparison with Prior Works.

This work proposed a lightweight, feedback-driven adaptation system that integrates a pre-trained ML model with pre-defined policy constraints as illustrated in Fig. 1. The dataset which used to train the model is generated based on the FMCW radar equations to guide offline training. Adaptation policies are defined to exploit spatial-temporal consistency, which reduces the reliance on unsupervised learning, enabling a low-latency and reliable adaptation [9]. We verified our system through two simulation scenarios and validated it with a small-scale experiment by embedding our adaptation framework on the Texas Instruments AWR1843 mmWave sensing module [10]. The results demonstrate that the system can accurately detect targets even during dynamic configuration updates in response to changing environments and achieves $2.7\times$ energy efficiency.

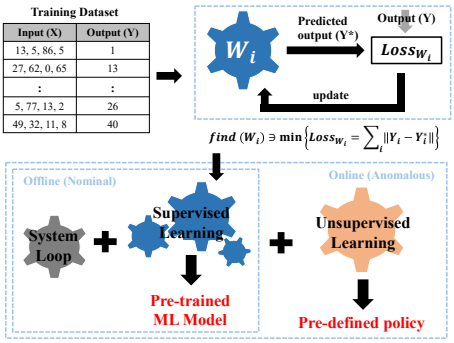


Fig. 1. Hybrid adaptation concept, uses a pre-trained ML model for offline and a pre-defined policy as a lightweight alternative to online learning.

II. Adaptation System Architecture

Fig. 2 illustrates the architecture of the proposed system, which dynamically adapts sensor configurations in real time. The closed-loop begins with a simplified RF front-end model—including TX nonlinearity and RX thermal noise—used in simulation and later replaced by a TI sensor module for validation.

The adaptation loop aims to maintain stable target information while reducing energy consumption. It extracts range, velocity, and noise from the range-Doppler spectrogram, then predicts future target positions. Errors between past predictions and current detections are used for correction. Large errors trigger baseline resets, while smaller ones adapt configurations via a pre-trained ML model.

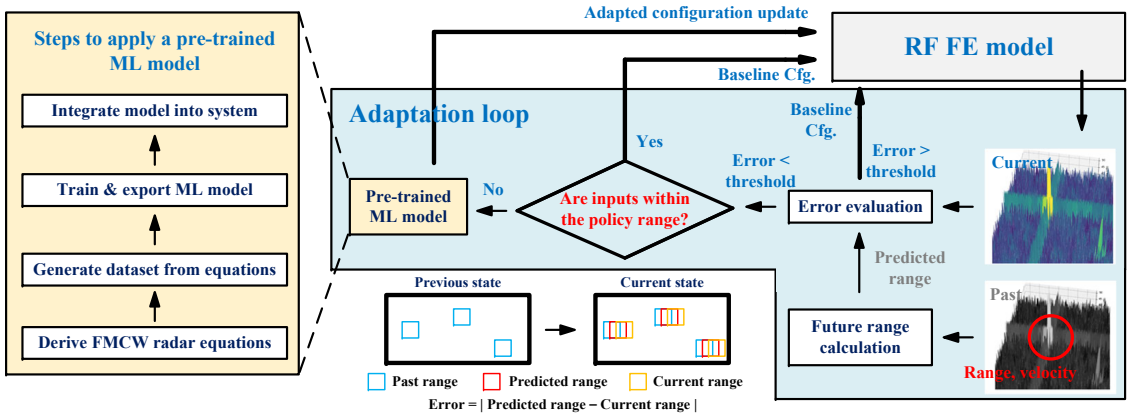


Fig. 2. Overview of FMCW radar sensor adaptation architecture. Range and velocity from Range-Doppler spectrograms guide configuration updates. A pre-trained ML model is used when policy constraints are met; otherwise, baseline settings are applied. The model is trained offline using radar-equation-based data for low-latency adaptation.

TABLE II
EQUATIONS FOR ADAPTABLE RADAR CONFIGURATIONS

Configuration	Equation
T_{chirp}	$S = \frac{B}{T_{\text{chirp}}}, \quad f_b = \frac{2RS}{c}, \quad \Delta R = \frac{c}{2B}$
N_{pulses}	$\Delta v = \frac{\lambda}{2T_{\text{coh}}}, \quad T_{\text{coh}} = N_{\text{pulses}} \cdot T_{\text{chirp}}$
P_{TX}	$P_{\text{TX}} = P_{\text{cons,TX}} \cdot \eta$
G_{RX}	$P_{\text{RX}} = P_{\text{TX}} G_{\text{TX}} G_{\text{RX}} \left(\frac{\lambda}{4\pi R}\right)^2 \sigma$
G_{BB}	$V_{\text{BB}} = G_{\text{BB}} \cdot V_{\text{RX}}$

TABLE III
ADAPTED CONFIGURATIONS (SIMULATION AND VALIDATION)

Configuration	Baseline	Range	Step
T_{chirp} (μs)	40	10 to 40	10
N_{pulses}	256	16 to 256	Factor of 2
P_{TX} (dBm)	12	6 to 14	1
G_{RX} (dB)	24	12 to 24	2
G_{BB} (dB)	24	12 to 24	2

The ML model takes target features (range, velocity, SNR) as input and outputs radar configurations for the next frame based on FMCW equations which are summarized in Table II, where S is the chirp slope, c is the speed of light, $P_{\text{cons,TX}}$ is the power consumption of the TX block, η is TX efficiency, and σ is the radar cross-section of the target. Energy consumption serves as the cost function, ensuring configurations are chosen to minimize power usage. Extreme cases are handled by pre-defined policies. For example, if a target is very close, adaptation is halted and a high range-resolution configurations is applied, even at the cost of reduced maximum unambiguous range. These policies can be customized depending on the application.

The baseline configurations—T_{chirp}, N_{pulses}, P_{TX}, G_{RX}, G_{BB}—along with their adaptation ranges and step sizes, are summarized in Table III. A typical mmWave radar frame lasts 40 ms. With 256 chirps (40 μs each, 8.5 μs gaps), chirp operations take around 12 ms, leaving 28 ms for synchronization and processing. Considering this timing constraint, we developed an adaptation loop that operates within a few ms without interfering with the sensing tasks.

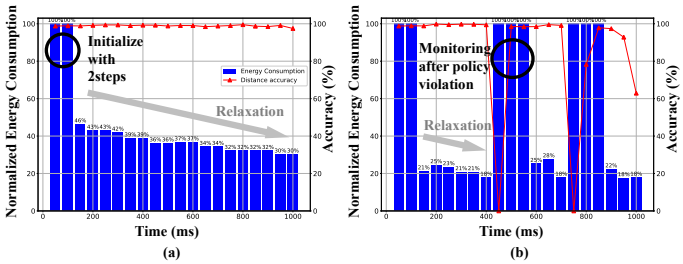
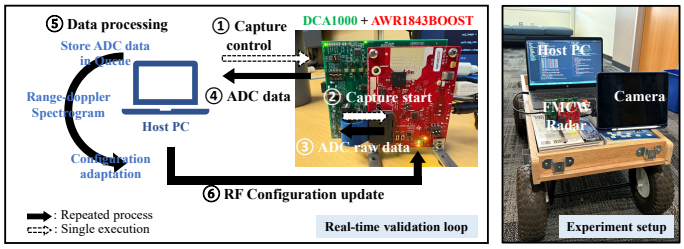


Fig. 3. Normalized energy consumption and detection accuracy in simulation. (a) The first scenario demonstrates a gradual reduction in energy consumption while maintaining high accuracy. (b) The second scenario introduces emergency cases to evaluate the system's reliability.

III. SIMULATION OF PROPOSED SYSTEM

Energy consumption ($E = P \cdot T = (\alpha G_{RX} + \beta G_{BB} + \gamma \cdot 10^{0.1 P_{TX}}) \cdot (\frac{(8.5e^{-3} + T_{Chirp}) \cdot N_{pulses}}{256} + 28)/40$) is modeled as the product of total power P and the active time T. The power term is decomposed into contributions from the RX chain and power amplifier, each weighted by coefficients α, β, γ to reflect their relative power consumption trends. The time term captures both frame-dependent activity and system overhead. LNA typically consumes 20 mW with RX gains up to 20 dB, motivating the choice of α. Baseband amplifiers consume 8 mW with gains of 20 dB, justifying a smaller β than α. PA efficiency is typically 30%, implying that a P_{TX} of 10 dBm yields a supply draw of 25 mW; γ accounts for this nonlinearity [11], [12].

Two scenarios are defined to verify the operation of the proposed system in terms of energy efficiency and reliability. In the first scenario, three objects are generated, each assigned a different range and velocity. Fig. 3(a) presents the results of the first scenario simulations. The first two steps involve evaluating whether adaptation is feasible by computing the error between the detected range and the predicted range. Once the error is determined to be lower than a threshold, the system adaptively updates its configurations to reduce energy consumption while maintaining high accuracy.



(a) (b)

Fig. 4. (a) Validation loop for real-time adaptation system using DCA1000 and AWR1843 FMCW Radar sensor. (b) Small-scale experimental setup.

In the second scenario, random target generation is introduced to evaluate the reliability of the adaptation system and the results shown in Fig. 3(b). When a policy violation occurs or a high error is evaluated, the system stops adaptation and reverts to the baseline configuration to ensure safety. After monitoring three steps for verifying the safety, the adaptation process is reinitialized to reduce energy consumption.

IV. Validation of Proposed System

Validation was conducted through a small-scale experiment in which DCA1000 and AWR1843 were connected to a host PC and mounted on a mobile platform. The validation setup is shown in Fig. 4(a), with a real-time raw ADC data handling implementation adapted from [13] as a baseline.

Fig. 4(b) illustrates a mounted validation setup on a mobile platform, enabling testing in dynamic environments. An integrated camera is used with the mmWave sensor module to establish a baseline for detection. The validation framework runs on a 12th Gen Intel(R) Core(TM) i5 CPU.

Fig. 5 demonstrates the real-world performance of the embedded adaptation system. The left panel shows a camera image, while the right panel presents mmWave radar sensor outputs including Range-Doppler and Range-Angle maps, along with the corresponding normalized energy consumption (E_{cons}) profile, consistent with the energy model used in the simulation. Bounding boxes in the camera image indicate the physical locations of three targets, and these correspond precisely to the reflections observed in Range Doppler/Angle maps, confirming the detection process. The E_{cons} plot demonstrates continuous adaptation over time, ensuring robust energy savings in dynamic environments.

Fig. 6(a) shows the latency of each component in the loop over 100 frames. Most of the delay occurs during the configuration update stage, due to communication latency between the host PC and the AWR1843. Adaptation latency ranges between 4 and 10 ms, and signal processing contributes the least. The average total loop delay is 34.3 ms, which is below the 40 ms single frame duration. This was sufficient under the current environment (64 chirps), but more dynamic or high-resolution scenarios (e.g., 256 chirps) may require a longer frame duration.

Fig. 6(b) presents the average E_{cons} of 100 frames, demonstrating a nearly 2.7× improvement in energy efficiency compared to the baseline in small-scale experiment. The results

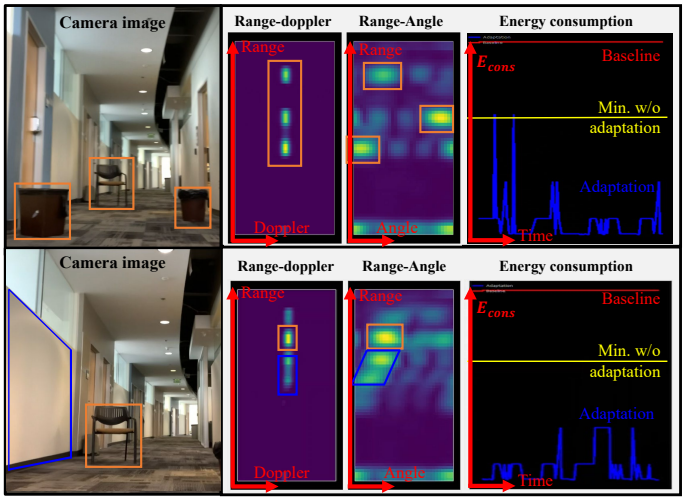


Fig. 5. Experimens with three objects in a hall using a mobile platform show that Range-Doppler/Angle maps confirm detection, while adaptive configuration reduces energy consumption compared to baseline.

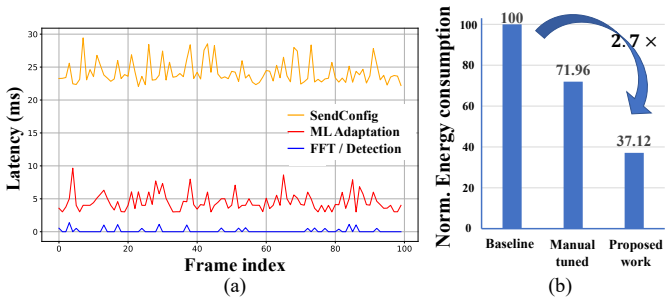


Fig. 6. Quantitative results of experiment. (a) Timing breakdown of each component in the real-time loop over 100 frames. (b) E_{cons} comparison between baseline, manually minimized configuration without adaptation, and the proposed adaptive approach.

validate the system's applicability in real-world scenarios, achieving low latency and high energy efficiency.

V. Conclusion and Discussion

This work presents real-time front-end adaptation framework for a mmWave radar sensor that uses a pre-trained ML model and pre-defined policy. By dynamically adapting the front-end based on environmental conditions within the mmWave sensor's time budget, the system achieves a 2.7× energy efficiency with robust accuracy. These results highlight the potential as a foundation for real-time front-end adaptive sensing. Although effective, the approach relies on fixed policies and threshold-based error, which may limit adaptability in unpredictable environments. Future work will focus on enhancing adaptation strategy for various conditions.

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